

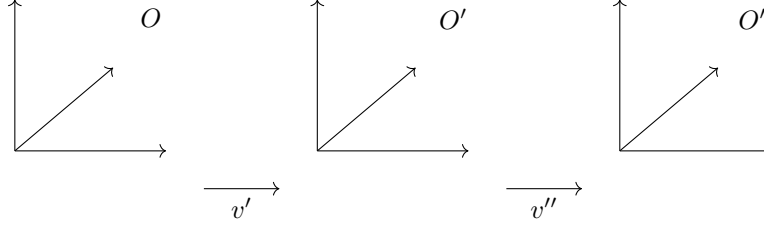
Chapter 0 Solutions

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RXZ

Solutions

1. Velocity Addition Formula



The composition of two Lorentz transformations is another Lorentz transformation. In other words, if we perform the transformation twice, it should still look like a Lorentz transformation. Since we are only boosting in the x-direction, we can simplify to dealing with 2×2 matrices. Boosting with v' and then with v'' means

$$\begin{pmatrix} ct'' \\ x'' \end{pmatrix} = \begin{pmatrix} \gamma'' & -\beta''\gamma'' \\ -\beta''\gamma'' & \gamma'' \end{pmatrix} \begin{pmatrix} \gamma' & -\beta'\gamma' \\ -\beta'\gamma' & \gamma' \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \\ = \begin{pmatrix} \gamma'\gamma''(1 + \beta'\beta'') & -\gamma'\gamma''(\beta' + \beta'') \\ -\gamma'\gamma''(\beta' + \beta'') & \gamma'\gamma''(1 + \beta'\beta'') \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}.$$

The resultant transformation should match

$$\begin{pmatrix} \gamma_u & -\beta_u\gamma_u \\ -\beta_u\gamma_u & \gamma_u \end{pmatrix}$$

where $u = c\beta_u$ is the velocity of O'' relative to O' . We want $\gamma'\gamma''(1 + \beta'\beta'') = \gamma_u = 1/\sqrt{1 - \beta_u^2}$. With some clever algebraic manipulation, we see that

$$\begin{aligned} \gamma'\gamma''(1 + \beta'\beta'') &= \frac{(1 + \beta'\beta'')}{\sqrt{(1 - \beta'^2)(1 - \beta''^2)}} \\ &= \sqrt{\frac{1 - \beta'^2 - \beta''^2 + \beta'^2\beta''^2}{(1 + \beta'\beta'')^2}}^{-1} \\ &= \sqrt{\frac{1 + 2\beta'\beta'' + \beta'^2\beta''^2 - 2\beta'\beta'' - \beta'^2 - \beta''^2}{(1 + \beta'\beta'')^2}}^{-1} \\ &= \sqrt{1 - \left(\frac{\beta' + \beta''}{1 + \beta'\beta''}\right)^2}^{-1} \end{aligned}$$

with

$$\beta_u = \frac{\beta' + \beta''}{1 + \beta'\beta''}.$$

This is known as the velocity addition formula.

2. Bianchi Identity

$F^{\mu\nu}$ is antisymmetric. This easily reduces the Bianchi identity to $\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0$.

We must lower the indices of $F^{\mu\nu}$. This is accomplished by the metric tensor.

$$F_{\mu\nu} = g_{\mu\alpha} F^{\alpha\beta} g_{\beta\nu}$$

$$= \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & -B_x & 0 \end{pmatrix}$$

Setting $\alpha = 2$, $\beta = 3$, and $\gamma = 0$,

$$\begin{aligned} \partial_2 F_{30} + \partial_3 F_{02} + \partial_0 F_{23} &= 0 \\ -\partial_y(E_z/c) + \partial_z(E_y/c) - \frac{1}{c}\partial_t B_x &= 0 \\ \partial_y E_z - \partial_z E_y &= -\partial_t B_x \\ (\nabla \times \vec{E})_x &= -\partial_t \vec{B}_x \end{aligned}$$

For the y th component, set $\alpha = 3$, $\beta = 1$, and $\gamma = 0$. For the z th component, set $\alpha = 1$, $\beta = 2$, and $\gamma = 0$.

3. The Dirac Delta

$$\begin{aligned} \tilde{f}(\omega) &= \int dt e^{-a|t|} e^{i\omega t} \\ &= \int_0^\infty dt e^{(-a+i\omega)t} + \int_{-\infty}^0 dt e^{(a+i\omega)t} \end{aligned}$$

The oscillatory part does not matter compared to the exponential behavior at large t , both positive and negative.

$$\begin{aligned} &= \frac{1}{a - i\omega} + \frac{1}{a + i\omega} \\ &= \frac{2a}{a^2 + \omega^2} \end{aligned}$$

Normalizing the Lorentzian:

$$\begin{aligned} C \int d\omega \frac{2a}{a^2 + \omega^2} &= 1 \\ 2C \tan^{-1}\left(\frac{\omega}{a}\right) \Big|_{-\infty}^{\infty} &= 1 \\ C &= \frac{1}{2\pi}. \end{aligned}$$

Thus,

$$\lim_{a \rightarrow 0} \frac{1}{\pi} \frac{a}{a^2 + \omega^2} \equiv \delta(\omega) = \frac{1}{2\pi} \int dt e^{i\omega t}.$$